

FINAL PROJECT REPORT

Global AI Adoption in Education

Data Analytics with Tableau

Project Technologies: PostgreSQL, DBeaver, Tableau Public, Python, Flask, HTML/CSS/JavaScript

Project Type: Data Analytics, Visualization and Web Integration

Note: The current implementation uses a 30-record synthetic/sample dataset prepared for project demonstration and analysis. It should not be presented as verified real-world global statistics.

1. Introduction

1.1. Project Overview

The project, **Global AI Adoption in Education**, presents an end-to-end data analytics solution for exploring artificial intelligence adoption and usage in education. The project combines database storage, data preparation, interactive Tableau visualizations, a dashboard, a Tableau Story, Tableau Public publishing, and a Flask-based web interface.

1.2. Objectives

The main objectives are to organize education-related AI adoption data, prepare the data for analysis, develop meaningful visualizations, build an interactive dashboard, communicate findings through a Story, test dashboard functionality, publish the analytical work through Tableau Public, and integrate the published views into a Flask web application.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Educational stakeholders need an accessible way to explore patterns in AI adoption, AI usage, infrastructure, impact, user type and barriers. The project addresses this need by transforming structured data into interactive analytical views and a web-accessible presentation.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a pipeline in which data is stored in PostgreSQL, prepared for Tableau analysis, represented through eight visualizations, combined into an interactive dashboard and Story, published on Tableau Public, and embedded into a Flask web application.

2.3. Initial Project Planning

The implementation was organized into database/data preparation, visualization, dashboard and Story creation, performance testing, Tableau Public publication, and Flask web integration. The final deliverable preserves the Tableau workbook and Flask source structure for reproducibility.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The project uses a structured 30-record sample dataset stored in PostgreSQL. The dataset includes fields covering country, region, year, education level, user type, AI tool, adoption rate, usage hours, student and educator counts, infrastructure score, AI impact score and barrier.

3.2. Data Quality Report

The project dataset was organized into a PostgreSQL table and connected to Tableau. The available implementation was tested through the resulting Tableau visualizations and dashboard. Because the dataset is synthetic/sample data, conclusions are intended for demonstration rather than validation of real-world global trends.

3.3. Data Exploration and Preprocessing

The data was connected from PostgreSQL to Tableau and prepared as a Tableau Extract for analysis. Fields were used to support comparisons by country, region, year, education level, user type, AI tool and barrier. Two calculated fields were used: Adoption Category and Impact Category, each grouping values into High, Medium and Low categories.

4. Data Visualization

4.1. Framing Business Questions

The analysis focuses on questions such as: How does AI adoption vary by country and year? How does AI usage differ by education level and user type? Which AI tools show higher impact? How does infrastructure vary by region? What barriers are associated with adoption?

4.2. Developing Visualizations

Eight visualizations were developed: AI Adoption Rate by Country; AI Adoption Rate Trend by Year; Average AI Usage Hours by Education Level; AI Impact Score by AI Tool; AI Adoption Rate by User Type; Infrastructure Score by Region; AI Adoption Rate by Barrier; and Student vs Educator AI Usage & Adoption.

5. Dashboard

5.1. Dashboard Design File

The dashboard, **Global AI Adoption Dashboard**, combines the project visualizations into a single interactive analytical interface. Filters are provided for Year, Country, Region and User Type, enabling users to focus the analysis on selected portions of the sample dataset. The dashboard was published through Tableau Public.

6. Report

6.1. Story Design File

A Tableau Story was developed to communicate the analysis sequentially. The published Story contains the project's analytical views, including an Executive Overview, geographic/adoption analysis, usage and education analysis, AI impact/tool analysis, and student/teacher comparison views. The Story is available through Tableau Public.

7. Performance Testing

7.1. Utilization of Data Filters

Four dashboard filter dimensions were implemented and tested: Year, Country, Region and User Type. The filters allow interactive exploration of the dashboard.

7.2. Number of Calculation Fields

Two calculated fields were implemented: Adoption Category and Impact Category.

7.3. Number of Visualizations

The project contains eight visualizations combined into the dashboard and used as analytical views in the Story.

7.4. Web Integration Testing

The Tableau Public Dashboard and Story were embedded in a Flask application using HTML iframe elements. The Flask application was tested locally at <http://127.0.0.1:5000>, and Dashboard/Story navigation was verified.

8. Conclusion/Observation

8.1. Conclusion

The project demonstrates a complete analytics workflow from PostgreSQL data storage through Tableau visualization and Story creation to Tableau Public publication and Flask web integration. The resulting system provides an interactive way to explore AI adoption-related measures in education.

8.2. Observation

The dashboard and Story make it easier to compare adoption, usage, infrastructure, impact and barriers across the available sample records. Since the dataset is synthetic/sample data, observations should be interpreted as demonstrations of analytical functionality rather than factual claims about global education systems.

9. Future Scope

9.1. Future Enhancements

Future work could incorporate larger and verified real-world datasets, automated data refresh, additional countries and institutions, predictive analytics, machine-learning-based adoption forecasting, more advanced Tableau parameters, user authentication, and cloud deployment of the Flask application.

10. Appendix

10.1. Source Code (if any)

The Flask source is organized as follows:

Global_AI_Flask/

■■■ app.py

■■■ templates/

■■■ index.html

The HTML page embeds the Tableau Public Dashboard and Story and provides Dashboard/Story navigation.

10.2. GitHub & Project Demo Link

GitHub repository: *To be added after repository creation.*

Tableau Public Dashboard:

https://public.tableau.com/views/Global_AI_Adoption_Public/GlobalAIAdoptioninEducation

Tableau Public Story: https://public.tableau.com/views/Global_AI_Adoption_Public/ExecutiveOverview

Local Flask demo: <http://127.0.0.1:5000> (local development URL; replace with a deployed URL if the application is hosted online).

Project Implementation Summary

Component	Status
PostgreSQL data storage	Completed
Data preparation / Tableau Extract	Completed
8 visualizations	Completed
Dashboard	Completed
Tableau Story	Completed
Performance testing	Completed
Tableau Public publication	Completed
Flask web integration	Completed

GitHub repository	To be created / added
Final report	Completed